

Souhaieb Elamri

Electrical Engineering Student | Computer Vision and Industrial AI for Automation

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PROFILE

Highly motivated Electrical Engineering student at ENSIT (pursuing a concurrent Master's in Advanced Robotics & AI). Building a strong technical foundation in computer vision, embedded systems, and industrial automation through hands-on projects. Seeking a final-year engineering internship (PFE) starting February 2027 to deploy AI solutions in a real-world industrial environment.

PROFESSIONAL EXPERIENCE

Summer Intern — WIC MIC GROUP06/2026 – 08/2026 | Bizerte, Tunisia

Tools: Python, YOLO11, YOLO11-Pose, DocTR, SVM, CatVTON, OpenCV, FastAPI, Streamlit, WebRTC, Docker, SQL Server

- Automated production ticket digitization using YOLO11, SVM, and FastAPI, completely eliminating manual data entry.
- Built a real-time virtual garment try-on platform (CatVTON, WebRTC) utilizing ArUco markers for centimeter-level spatial accuracy.

Master's Thesis Intern — WIC MIC GROUP02/2025 – 05/2025 | Bizerte, Tunisia

Tools: Python, OpenCV, PyTorch, YOLO11-Pose, CUDA, Docker, SQL Server

- Engineered a real-time AI garment inspection system, managing everything from massive data annotation to OpenCV multi-camera calibration, achieving ±0.3 cm precision.

Bachelor's Graduation Project — HYCO Industries - Metal Construction02/2023 – 05/2023 | Nabeul, Tunisia

- Retrofitted a manual metal fabrication machine into an automated multi-axis CNC system, integrating stepper motor drivers and configuring Mach3 control software via a Raspberry Pi interface.

Technical Intern — CMR Tunisia Ship Repairs01/2023 – 02/2023 | Bizerte, Tunisia

- Automated a marine basin control system by developing a PLC/HMI architecture using Siemens PLCs, Ladder Logic, and TIA Portal for real-time pump, valve, and water-level control.

EDUCATION

National Engineering Diploma in Electrical Engineering2025 – 2027 (expected)

National Higher School of Engineers of Tunis (ENSIT)

Master's Degree in Advanced Robotics and Artificial Intelligence2023 – 2025

Higher Institute of Technological Studies of Bizerte

Bachelor's Degree in Electrical Engineering – Industrial Automation2020 – 2023

Higher Institute of Technological Studies of Nabeul

ACADEMIC PROJECTS

Smart Embedded System for Wildfire Prevention2026 – Present

- Designed an embedded environmental monitoring node using a Raspberry Pi. Integrated SPI/I2C sensors to process 4 environmental metrics (CO2, moisture, temperature, wind) and transmit low-latency hazard alerts.

Intelligent Mobile Robot2025 – 2026

- Designed a mobile robotics platform using ESP8266 and Node-RED. Engineered PWM motor controls and programmed I2C/ADC sensor data acquisition to provide real-time IoT telemetry.

TECHNICAL SKILLS

Embedded & Industrial: Raspberry Pi, Arduino, ESP32, STM32, FPGA, VHDL, TIA Portal, PLC, CNC, LabVIEW

Generative AI & NLP: RAG, Local LLMs (Llama 3 & Mistral 7B via Ollama)

Tools & Languages: Python, C, MATLAB/Simulink, SolidWorks, FastAPI, SQL Server, Docker, Git

Computer Vision & Inference: OpenCV, YOLO, PyTorch, Object Detection, Image Segmentation, Pose Estimation, Multi-Camera Calibration, ArUco, Pyzbar, OCR (DocTR), ONNX, TensorRT, CUDA

CERTIFICATIONS

C++ For C Programmers, Part A (UC Santa Cruz) — 02/2025

Six Sigma Yellow Belt — 01/2025

Microsoft Certified: Security, Compliance, and Identity Fundamentals (SC-900) — 11/2024

Convolutional Neural Networks (DeepLearning.AI) — 06/2024

Microsoft Certified: Azure AI Fundamentals (AI-900) — 05/2024

Neural Networks and Deep Learning (DeepLearning.AI) — 03/2024

Supervised Machine Learning: Regression and Classification (Stanford Online) — 01/2024

LANGUAGES

French — Fluent | English — Fluent | Arabic — Native